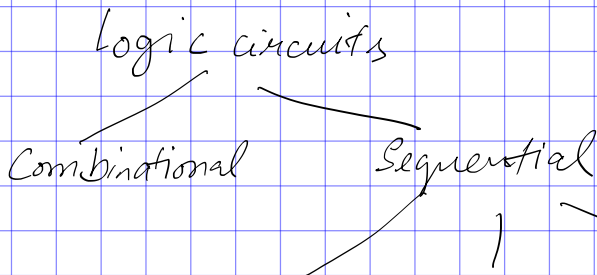
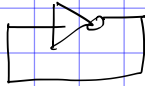


# Sequential Circuits

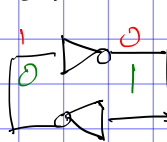


monostable

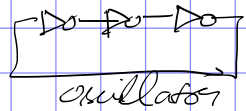
$V_{in} = V_{out} = V_{th}$



Bistable

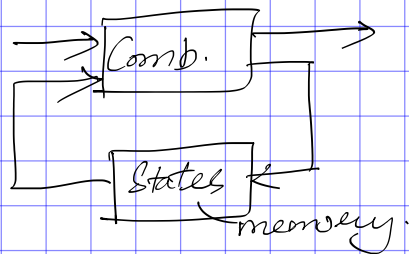


Astable

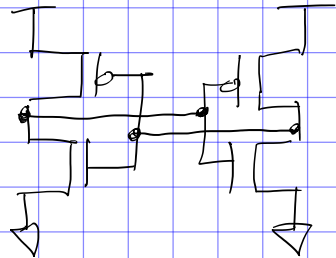
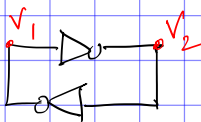


oscillator

A typical FSM

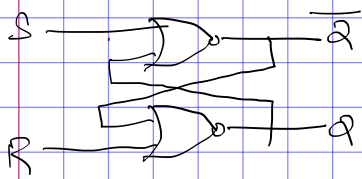


In all the memories we'll be looking into we will find the back to back connected inverter topology.

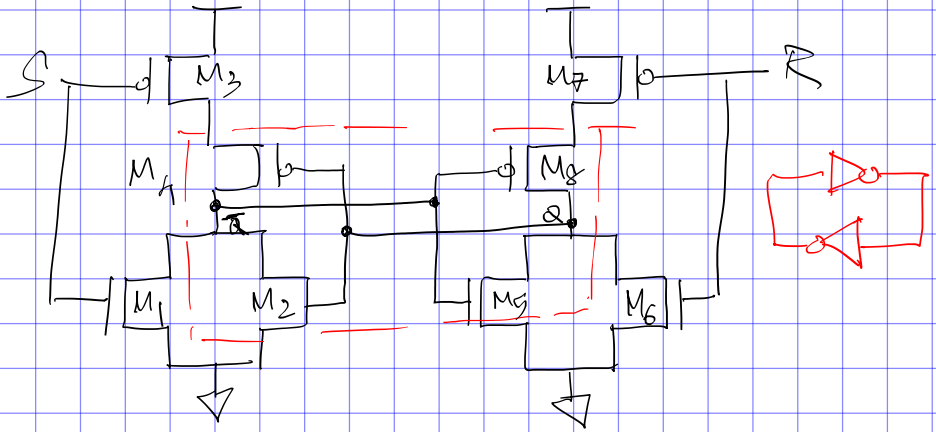


Set Reset Latch

| S | R | $Q_{n+1}$ | $\overline{Q}_{n+1}$ | Action  |
|---|---|-----------|----------------------|---------|
| 0 | 0 | $Q_n$     | $\overline{Q}_n$     | Hold    |
| 0 | 1 | 0         | 1                    | Reset   |
| 1 | 0 | 1         | 0                    | Set     |
| 1 | 1 | -         | -                    | Invalid |



NOR implementation of SR latch.

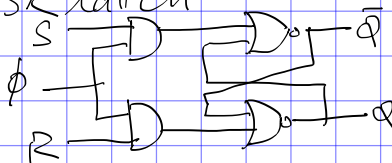


S  $V_{DD}$   
R 0

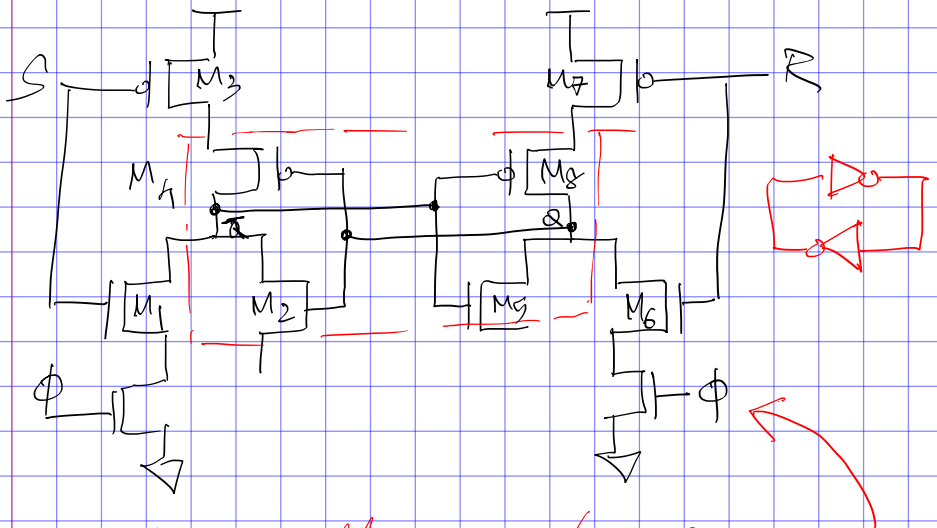
$M_3$  &  $M_6$  are "off",  $M_1$  &  $M_7$  are "on".  
Assume  $\bar{Q} = V_{DD}$  &  $Q = 0$  before:  
 $\Rightarrow M_4$  &  $M_1$  are "on" but no pull up path to charge  $\bar{Q}$   
 $\Rightarrow \bar{Q}$  discharges via  $M_1$   
when  $\bar{Q}$  reaches  $V_{IL}$ ,  $M_8$  turns "on" and  $M_5$  approaches "OFF" state.  
 $\Rightarrow Q$  gets pulled up to  $V_{DD}$

Similar arguments can be made for other input & output combinations.

clocked SR latch

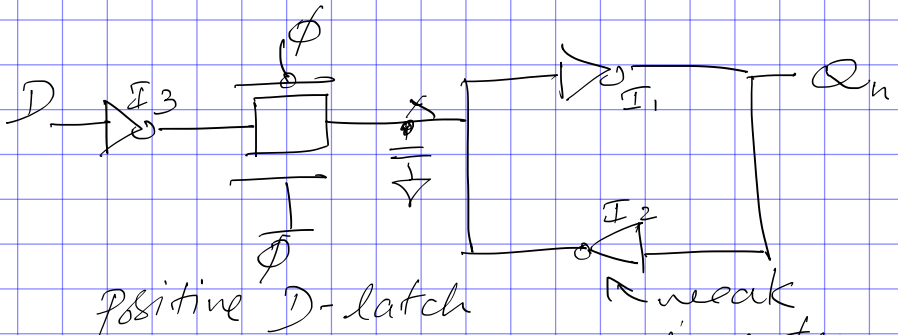


level triggered.  
Allows S, R to control Q during the positive level.

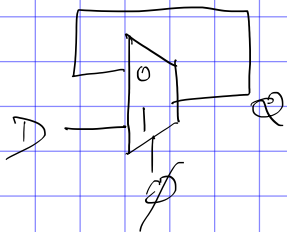


HW Why is adding the  $\phi$  to PDN sufficient?

D-latch



A weak inverter in the feedback path so that  $I_3$  can override  $I_2$  even when  $I_2$  is active.

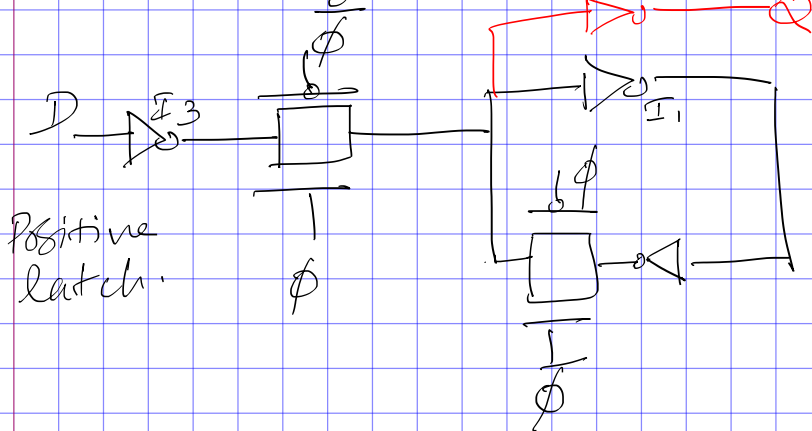


$$Q_{n+1} = \overline{\phi} Q_n + \phi D$$

1

Muxes are implemented using transmission gates.

To avoid the overhead of a weak inverter a transmission gate is used instead.



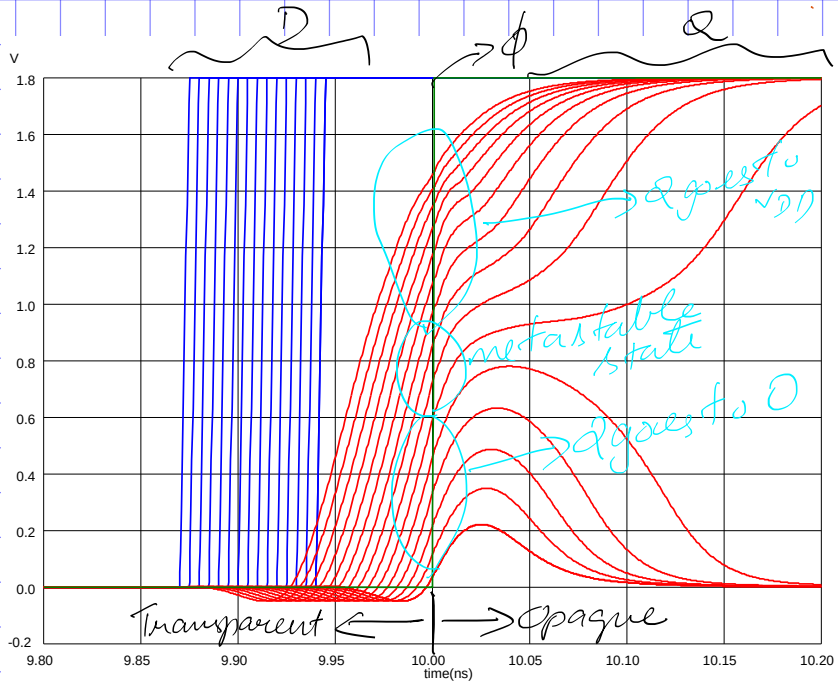
Positive latch.

To avoid loading the "memory" element with an external gate,  $Q$  is read using a separate inverter.

## Metastability

For voltages close to  $V_{th}$  back to back connected inverters take a while to reach either of the stable states.

This "sluggish" behaviour is termed metastability. This phenomenon is shown below for the negative D-latch. When the  $\phi$  input rises the latch transitions from transparent to opaque.



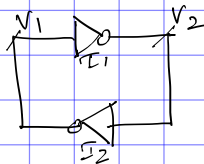
A rising  $D$  input transition is brought close to the clock edge.  $\phi$  and  $\bar{\phi}$  are assumed to transition at the same time and have a very small rise/fall time.

The values of  $Q$  during the  $\phi$  transition indicate the sampled state which decides whether  $Q$  will climb to  $V_{DD}$  or fall to 0. Annotated in the figure are the regions where  $Q$  goes to  $V_{DD}$  & 0.

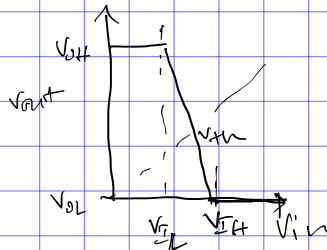
## Metastable state:

In the region in between  $Q$  takes a while to transition to either  $V_{DD}$  or  $0$ . We are not interested in this non-deterministic state and require the  $D$  to be captured properly. This gives rise to the "setup" time of a latch: The time before the clock transition by which the data has to be ready to be captured reliably by the latch.

## Simple model of metastability

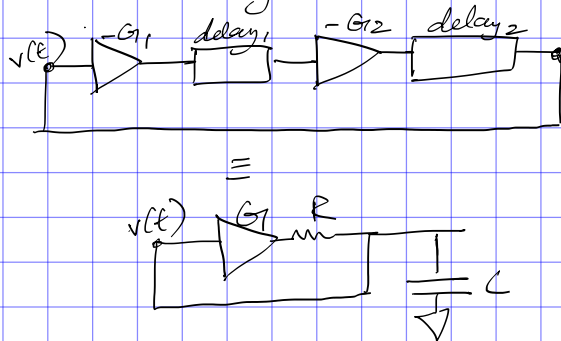


Consider an approximate version of the VTC of inverter



Let  $G_1$  be the mag. of gain of the inverter at  $V_{th}$ .

Assuming  $I_1, I_2$  have the same  $V_{th}$  and gain, the inverters can be modelled in small signal as



RC models ( $\text{delay}_1 + \text{delay}_2$ ) &  $G = G_1 \times G_2$

$v(t)$  is the incremental voltage above  $V_{th}$ .

KCL: 
$$\frac{G_1 v(t) - v(t)}{R} = C \times \frac{dv(t)}{dt}$$

$$v(t) = v(0) e^{\frac{t}{\tau}}, \text{ where } \tau = \frac{RC}{G_1 - 1}$$

Suppose we are interested in the time it takes for the inverters to reach  $V_{IH}$ .

$$v(t) = V_{IH} - V_{th}$$

$$t = (\ln(V_{IH} - V_{th}) - \ln(v(0))) \times \tau$$

If  $v(0) \rightarrow 0$ ,  $t \rightarrow \infty \Rightarrow$  as the sampled voltage gets closer to  $V_{th}$  it takes longer to reach one of the bistable states.

Measurement of setup time:

$t_{DQ}$  = time between  $D$  rise to  $Q$  rise

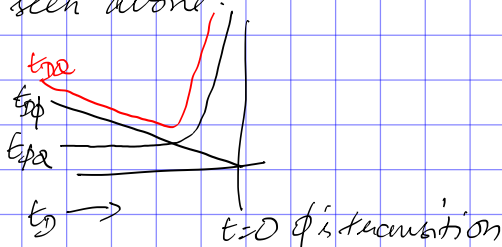
$t_{D\phi}$  = time between  $D$  rise to  $\phi$  rise

$t_{\phi Q}$  = time between  $\phi$  rise to  $Q$  rise.

$$t_{DQ} = t_{D\phi} + t_{\phi Q}$$

As  $D$ 's transition is brought closer to  $\phi$ 's transition the clock to  $Q$  delay  $t_{\phi Q}$  becomes larger due to metastability as seen above.

$\Rightarrow$



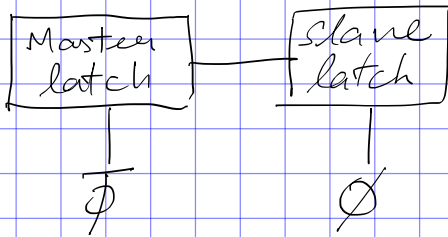
$t_{D\phi}$  decreases as  $D$  gets closer to  $Q$ .

$\Rightarrow t_{DQ}$  decreases and then increases.

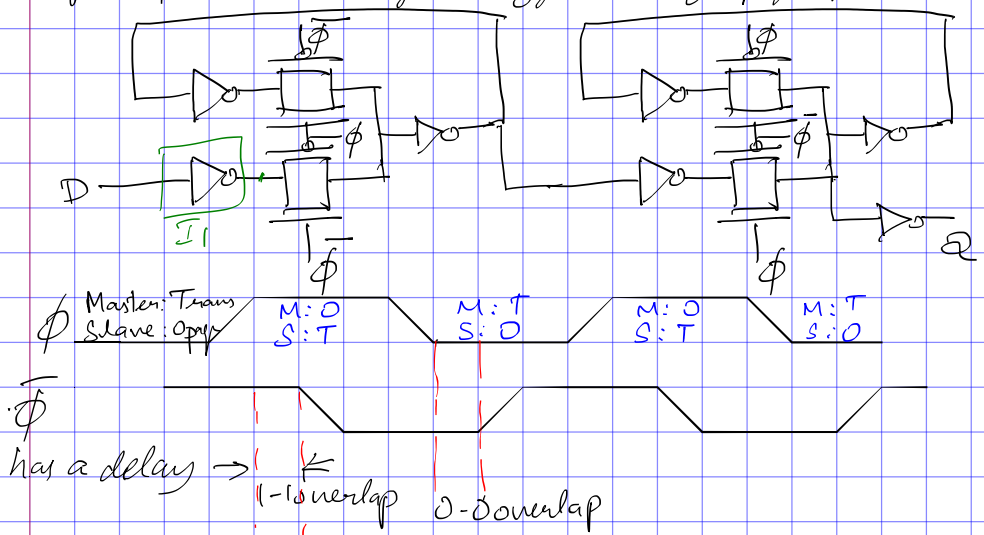
$(t_{setup}, t_{clock-Q})$  are  $(t_{D\phi}, t_{\phi Q})$  at which  $t_{DQ}$  is minimum.

# Flipflops - Edge triggered

Realised using two latches



The same analysis of setup holds good for the flip. Positive edge triggered flipflop.



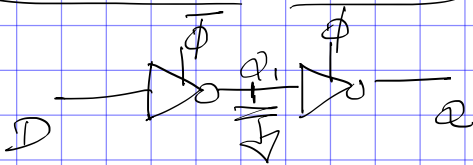
During 1-1 overlap the nMOS of all transmission gates are "on".  $\Rightarrow$  there is a direct path from D to Q. Any transition in D in this window can race through to Q. Since this happens after the positive edge, this introduces the hold time constraint: the time after the clock transition within which the input cannot transition.  $\Rightarrow t_{hold} \geq 1-1 \text{ overlap time}$ .



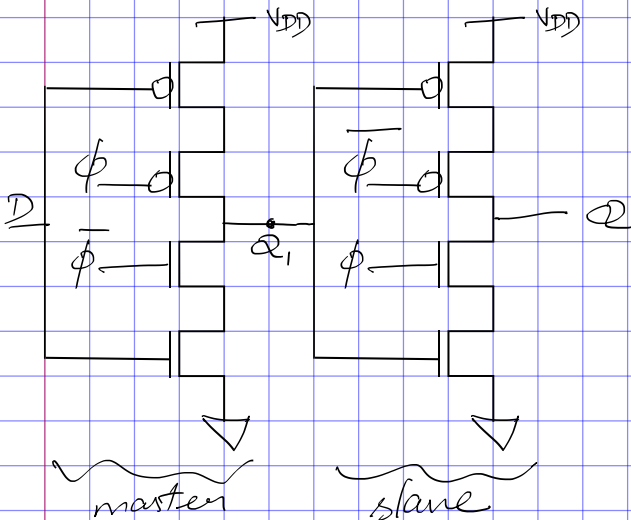
During  $\phi$ - $\phi$  overlap all the pMOS are "on" which once again leads to a race from  $D$  to  $Q$ . This happens in the non sampling edge of the flip-flop and hence cannot be overcome using a hold constraint. This imposes. The data at  $D$  cannot race through to  $Q$  if the delay of inverter  $T_i$  is larger than the  $\phi$ - $\phi$  overlap time.

To overcome this a dynamic  $D$ -flipflop is often used:

### Clocked CMOS - $C^2$ MOS

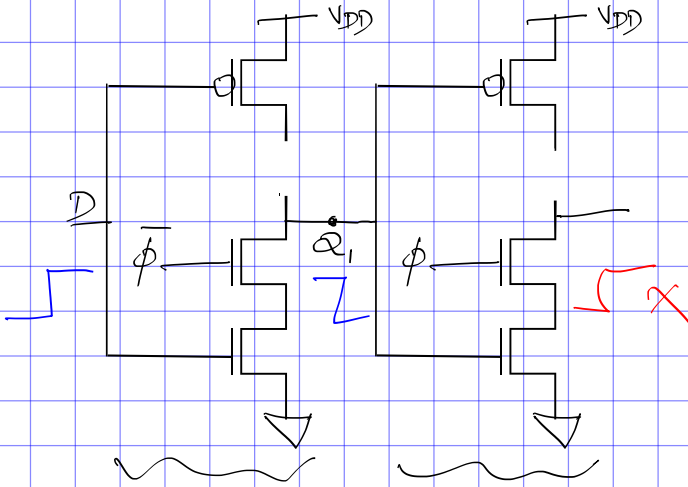


Built using back-to-back tri-state inverters.



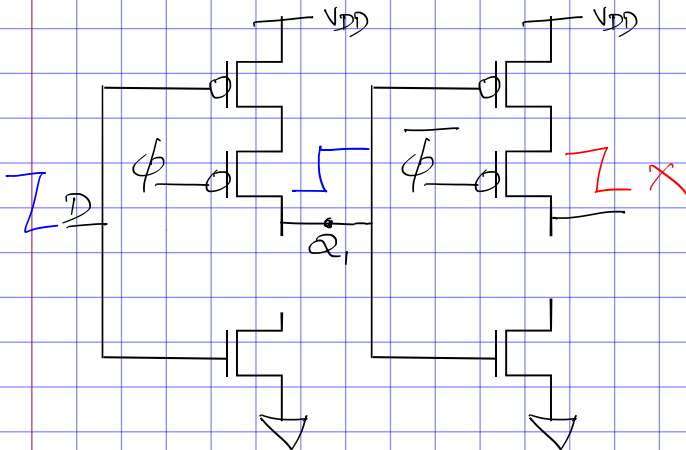
When the master latch turns "off" the charge in  $Q_1$  isn't connected to  $V_{DD}$  or  $\phi$  making this circuit dynamic.

During 1-1 overlap the flop behaves as



A rising transition at  $D$  can cause a falling transition at  $Q_1$ , but this cannot propagate further. But after the 1-1 overlap the slave turns transparent causing the transition to break through. This can be overcome using the hold time constraint.

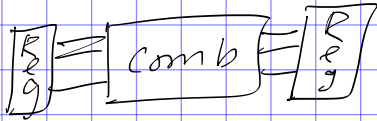
For 0-0 overlap



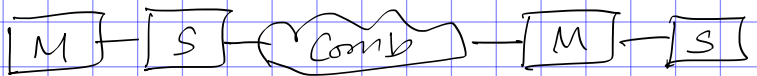
A falling transition at  $D$  can trigger rising at  $Q_1$ , and slave is "off" after 0-0 overlap thereby preventing transition from going through. This flip-flop has no race condition for 0-0 overlap.

## Time borrowing in latches

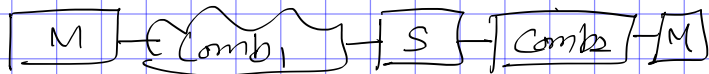
A typical combinational block looks like



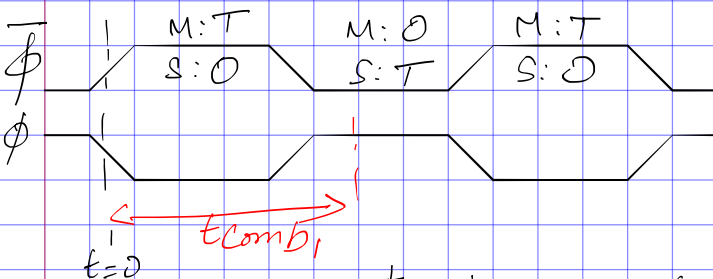
Registers/flip flops are composed of master/slave latches:



If we partition the combinational block into two such that



If the delay of  $\text{comb}_{1/2}$  is half the delay of  $\text{comb}$  then the migration of latches has no effect on the clock frequency.



Suppose  $t_{\text{comb}1} > t_{\text{comb}2}$  and  $t_{\text{comb}2} < t_{\text{comb}2}$

Since slave is transparent here the data from  $\text{comb}_1$  can be used by  $\text{comb}_2$  as long as  $\text{comb}_2$ .

We can continue this argument to comb<sub>2</sub>.

This results in the following inequality.

$$t_{\text{comb}_1} < T - t_{\text{setup}} \text{ where } T \text{ is the clock period}$$
$$t_{\text{comb}_2} + t_{\text{comb}_1} < \frac{3T}{2} - t_{\text{setup}}$$

$$t_{\text{comb}_3} + t_{\text{comb}_2} + t_{\text{comb}_1} < 2T - t_{\text{setup}}.$$

and so on.

This is the concept of time borrowing where a stage borrows time from the succeeding block.